INTRODUCTION

How and why do species change over time? In this activity, you'll explore real data that scientists collected to investigate this question. These data were taken from a population of birds called **finches**, before and after a drought in the Galápagos Islands.

In Part 1, you'll read about the finches and the scientists who studied them: Peter and Rosemary Grant. In Part 2, you'll analyze some of the Grants' data by calculating key statistics. In Parts 3 and 4, you'll use this data to construct graphs and perform statistical analyses (in particular, a *t* test) to make conclusions about natural selection. The concepts you'll learn apply not only to finches in the Galápagos but also to any organism in any environment.

MATERIALS

- access to *Data Explorer* or a spreadsheet program
- the finch data set (in *Data Explorer* or as a spreadsheet)
- (optional) calculator and graphing materials (colored pencils, rulers, graph paper, etc.)

PART 1: Learning about the Study

In 1973, Princeton University evolutionary biologists Peter and Rosemary Grant began studying the finches of the Galápagos archipelago, a group of islands about 600 miles off the coast of Ecuador. They collected thousands of measurements every year to track changes in the physical characteristics of finch populations over time. One of their major goals was to collect enough data to identify associations between environmental and evolutionary changes in finch populations.

For their study, the Grants focused on the medium ground finch (*Geospiza fortis*), a seed-eating species of finch on the island of Daphne Major. Every year, the Grants measured physical characteristics like **wing length**, **body mass**, **tarsus length** (the section of leg between the ankle and knee), and **beak depth** (a measure of beak size) for hundreds of individual medium ground finches (Figure 1). Small differences in these characteristics can be important for survival in different environments. In addition, these characteristics tend to vary widely within finch populations.

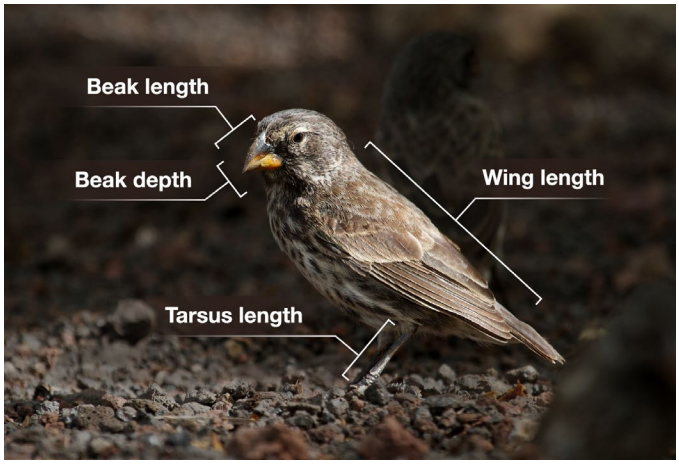


Figure 1. Diagram showing four physical characteristics that were measured in the finches.

In early 1977, a drought began on Daphne Major. This drought caused the type and abundance of food available to the finches to change rapidly. Medium ground finches prefer to eat the small, soft seeds of a bushy plant called chamaesyce (*Chamaesyce amplexicaulis*). But during the drought, chamaesyce seeds became very scarce. One of the only remaining food sources for the finches was the seeds of a plant called caltrop (*Tribulus cistoides*). Caltrop seeds are much larger and harder than chamaesyce seeds and are covered with pointy spines.

The drought lasted for 18 months and killed over 80% of the 1,200 medium ground finches on the island. The Grants were interested in determining whether there were any differences between the finches that survived the drought (**survivors**) and the finches that did not (**nonsurvivors**) — in particular, whether any physical characteristics were key to survival. To answer this question, the Grants compared the average values of different characteristics in the survivors to the average values of the same characteristics in the nonsurvivors. They then used statistical methods to determine whether the differences they found between the two groups were likely to have affected survival or just occurred by chance.

1. Describe the major environmental change on Daphne Major that took place in 1977.

2. Based on what you know so far, what differences (if any) do you think there were between the finches that survived this major environmental change and the finches that did not?

To see whether the differences you expected are supported by the data, you will explore one of the Grants' data sets. Open the finch data set. In *Data Explorer*, you can do this by clicking "Choose data to explore" on the landing page, selecting the "Finches in the Galápagos" data set, and then clicking the **"Data"** tab at the top.

This data set contains measurements of physical characteristics (body mass, wing length, tarsus length, and beak depth) from 100 medium ground finches on Daphne Major, all born between 1973 and 1976. The 50 finches labeled as nonsurvivors did not survive the drought and died in 1977. The 50 finches labeled as survivors survived the drought and were still alive in 1978.

PART 2: Calculating Descriptive Statistics

You will now calculate descriptive statistics for physical characteristics in this sample of finches. **Descriptive statistics** are numbers that summarize the features of the data and their distribution. Table 1 lists five types of descriptive statistics: mean, variance, standard deviation, standard error of the mean, and 95% confidence interval. As you'll see by doing the calculations below, these statistics are related to each other and can reveal information not only about the sample, but also the entire population from which the sample was taken.

Follow the steps below the table to complete the calculations. You can do the calculations by hand, in *Data Explorer* (under the **"Analyze"** tab at the top), or with another program. Make sure to record each calculation in Table 1. The variance calculations are filled in for you as an example.

Table 1. Descriptive statistics for four physical characteristics in a sample of 100 medium ground finches: 50 survivors and 50 nonsurvivors.

	Nonsurvivors				Survivors			
Descriptive statistics	Body mass (g)	Wing length (mm)	Tarsus length (mm)	Beak depth (mm)	Body mass (g)	Wing length (mm)	Tarsus length (mm)	Beak depth (mm)
Mean								
Variance (s^2)	1.842	5.181	0.701	0.775	3.087	5.448	0.735	0.709
Standard deviation (s)								
Standard error of the mean (SEM)								
95% confidence interval (CI)								

The first descriptive statistic for you to calculate is the **mean**, or average, of the sample.

3. Calculate the mean for each column in Table 1, and record your results (rounded to two decimal places) in the table.

The next descriptive statistic to calculate is the sample **standard deviation** (s), which measures the mean difference between each individual measurement and the mean of the sample. Standard deviation is a way to quantify how spread out a set of measurements is compared to the mean.

The sample standard deviation is the square root of the sample variance (s^2). You can use the variances shown in Table 2 to calculate the standard deviations.

4. Calculate the standard deviation for each column in Table 1, and record your results (rounded to two decimal places) in the table.

The next descriptive statistic to calculate is the **standard error of the mean (SEM)**. The SEM is the standard deviation of the means of multiple random samples taken from the same population. About 68% of the sample means are within one SEM of the population mean.

As a result, the SEM can be used to measure how well the mean of any random *sample* estimates the mean of the entire *population*. You can use the formula below to estimate the SEM:

$$\text{SEM} = \frac{s}{\sqrt{n}}$$

5. Calculate the standard error of the mean for each column in Table 1, and record your results (rounded to two decimal places) in the table.

The last descriptive statistic to calculate is the **95% confidence interval (CI)**. The 95% CI provides a range around the *sample* mean within which the mean of the *entire population* is likely to be found. (Specifically, there is a 95% chance that the 95% CI contains the mean of the entire population.) You can approximate the 95% CI as twice the SEM.

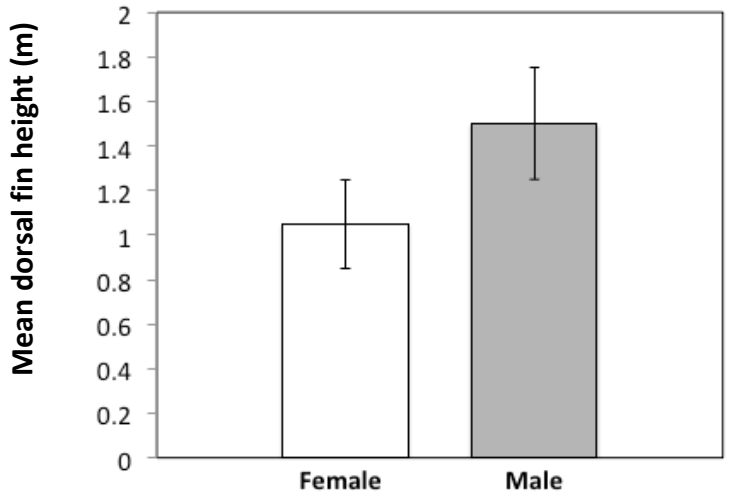
6. Calculate the 95% CI for each column in Table 1, and record your results (rounded to two decimal places) in the table.

PART 3: Graphing the Data

Let's compare the means you calculated for the two groups using graphs. A bar graph is often a good choice for comparing a single numerical value among different groups.

Look at the following example of a well-constructed bar graph. Note that it contains a title, labels for the axes (including units), and an explanation for what the error bars represent.

Mean dorsal fin height among male and female orcas



Mean dorsal fin height in meters (m) for 36 female and 36 male orcas (*Orcinus orca*). In this case, error bars represent 95% CIs.

7. Construct four bar graphs that compare the means of each physical characteristic (body mass, wing length, tarsus length, and beak depth) for survivors and nonsurvivors.
   - You can draw the graphs by hand on a separate sheet of graph paper, or you can make them on a computer using *Data Explorer* (under the **"Visualize"** tab at the top) or another program. Make sure to include all your graphs when submitting this handout.
   - For each graph, make sure to include a title, labels for the axes (including units), and error bars that represent 95% CIs.
8. In some cases, whether or not your error bars overlap can suggest whether the difference between the means is **statistically significant** — that is, due to actual differences between nonsurvivors and survivors instead of just happening by chance.
   a. You created error bars that represent 95% CIs. What does it suggest if these error bars do *not* overlap? What does it suggest if these error bars *do* overlap?

   b. For which of the four physical characteristics did your error bars *not* overlap?

9. Summarize the differences between nonsurvivors and survivors that you observed from your graphs. Make sure to consider each of the four physical characteristics and the error bars from the previous question.

PART 4: Calculating *t* Test Statistics

To confirm whether the differences you found in Part 3 are statistically significant, you will now do a statistical test. The appropriate statistical test for comparing two sample means is the **Student's *t* test**, also known as just the *t* test, which assesses whether the difference between the means of two samples (in this case, the 50-bird nonsurvivor and survivor samples) happened simply by chance and is *not* statistically significant. This would indicate that the means of the populations that the samples came from (in this case, the overall nonsurvivor and survivor groups) are the same.

This hypothesis — that the mean of the population from which the first sample came is *equal* to the mean of population from which the second sample came (so any differences in the sample means just happened by chance) — is called the **null hypothesis**. In this case, the null hypothesis will be that, for a given trait, the mean of the nonsurvivors equals the mean of the survivors.

First, you will calculate what is called the **observed *t* statistic** (t_{obs}). Then, you will compare the absolute value of t_{obs} to the **critical *t* statistic** (t_{crit}), a cutoff value that determines whether you can reject the null hypothesis.

- If $|t_{obs}| < t_{crit}$, you *cannot* reject the null hypothesis. In other words, the means of survivors and nonsurvivors are *not* significantly different.
- If $|t_{obs}| > t_{crit}$, you *can* reject the null hypothesis. In other words, the means of survivors and nonsurvivors *are* significantly different.

The value of t_{crit} is based on sample size. It represents the *t* statistic that could occur 5% of the time in that sample size if the null hypothesis were true. For example, each of your nonsurvivor and survivor samples had 50 birds. The t_{crit} for a sample size of 50 is 1.98. This means that, if the null hypothesis is true, we'd expect a $|t_{obs}|$ of 1.98 or greater to happen only 5% of the time.

10. Calculate t_{obs} to compare the means of each physical characteristic between survivors and nonsurvivors. You can calculate t_{obs} using *Data Explorer* (under the "**Analyze**" tab at the top), a spreadsheet program, a graphing calculator, or an online *t* test calculator.
    a. Record your results for $|t_{obs}|$ (the absolute value of t_{obs}) in the following table.

Characteristic	$\|t_{obs}\|$
Body mass	
Wing length	
Tarsus length	
Beak depth	

    b. For each characteristic, how does $|t_{obs}|$ compare to the t_{crit} of 1.98?

    c. For each characteristic, make a claim about whether the means of the survivors and nonsurvivors are significantly different. Support your claim with evidence from the *t* test.

11. Answer the following questions about the finches based on your analysis of the data.
    a. Which of the four physical characteristics do you think were important for the finches' survival during the drought? Explain your reasoning.

    b. Based only on these data, can you determine which characteristic was the *most* important for survival during the drought? If yes, explain how you know. If no, describe what other information you might need.

    c. Suggest a biological reason for the differences you found between the nonsurvivors and survivors.

EXTENSION: Evaluating Associations Between Variables

In the previous parts of this activity, you explored several physical characteristics of the finches. Some of these characteristics might be **associated**, or related to each other. For example, finches with bigger bodies might also generally have bigger beaks.

Let's explore the potential association between two of the characteristics: beak depth and wing length. One way to examine the association between two variables is by using a graph such as a scatterplot. A **scatterplot** uses dots to represent pairs of values for the variables.

1. Construct a scatterplot showing the beak depths and wing lengths of the *survivors* in the finch data set.
   - You can make the graph on a computer using *Data Explorer* or another program. Make sure to include all your graphs when submitting this handout.
   - Make sure to include a title and labels for the axes (including units).
2. Add a linear regression line, also known as a "trendline," to your graph in the program you are using. (In *Data Explorer*, you can do this by creating a scatterplot in the "Visualize" tab, then toggling on the linear regression line in the "Edit plots" window.)

In addition to using a graph to visualize associations, you can also calculate the **coefficient of determination** (r^2). This value represents the proportion of the variation in the *y* variable that is "explained" by the variation in the *x* variable. For example, if r^2 is 0.90, the *x* variable "explains" 90% of the variation in the *y* variable. Values of r^2 vary from 0 to 1, with larger values indicating a stronger association between the variables.

3. Calculate the **coefficient of determination** (r^2) for beak depth and wing length in the program you are using. In *Data Explorer*, you can do this by selecting "Linear regression" under the "Analysis" tab. Record your result below.

4. Based on your results, is there an association between beak depth and wing length in this group of finches? Why or why not?

5. Suggest a biological reason for the presence or absence of an association between beak depth and wing length in this group of finches.

6. Imagine that the abundance of small, soft seeds resumed after the drought ended in 1978. Predict what would happen to **body mass** in this finch population over later generations. Explain your answer. (*Hint*: Think about whether and how body mass could be associated with beak depth and wing length. You can make another graph or do additional calculations to check your expectations.)